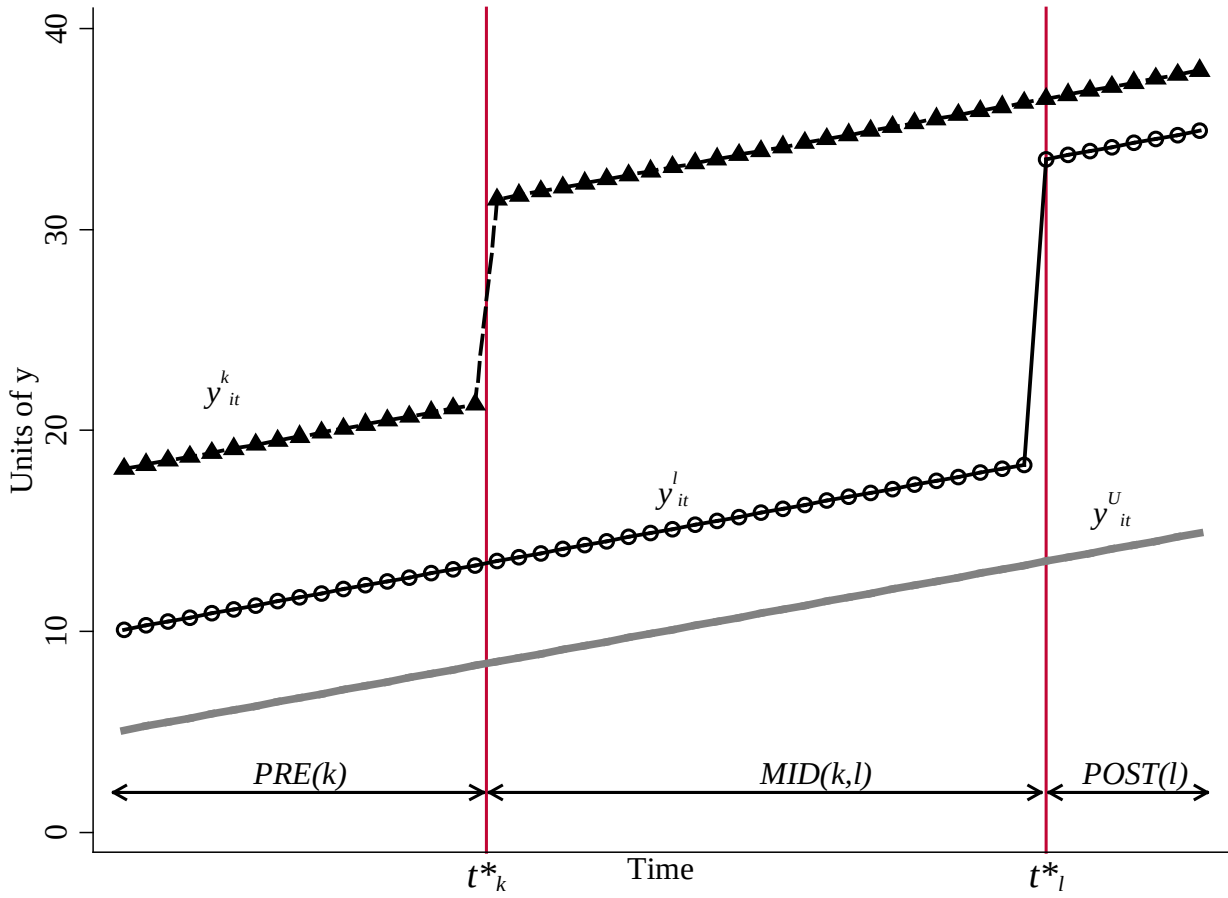
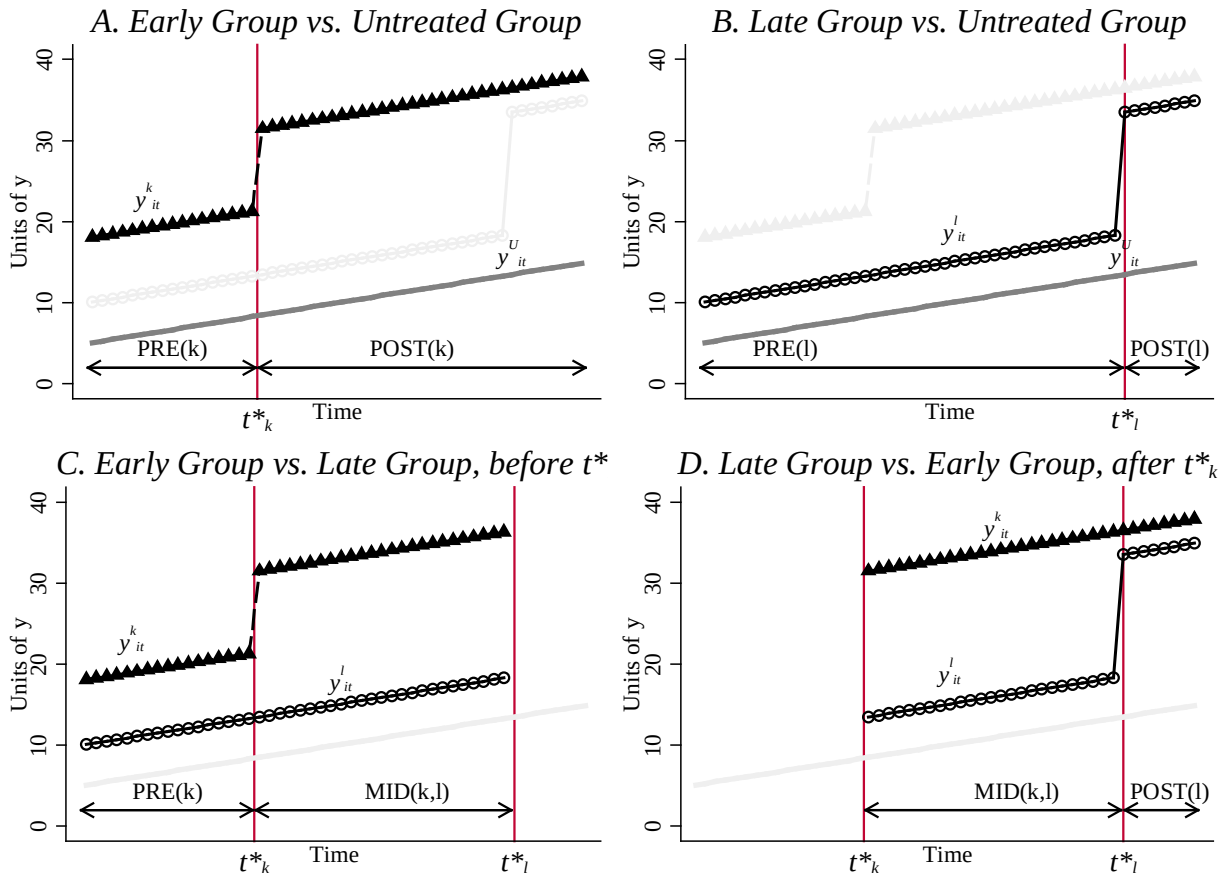


Figure 1. Difference-in-Differences with Variation in Treatment Timing: Three Groups



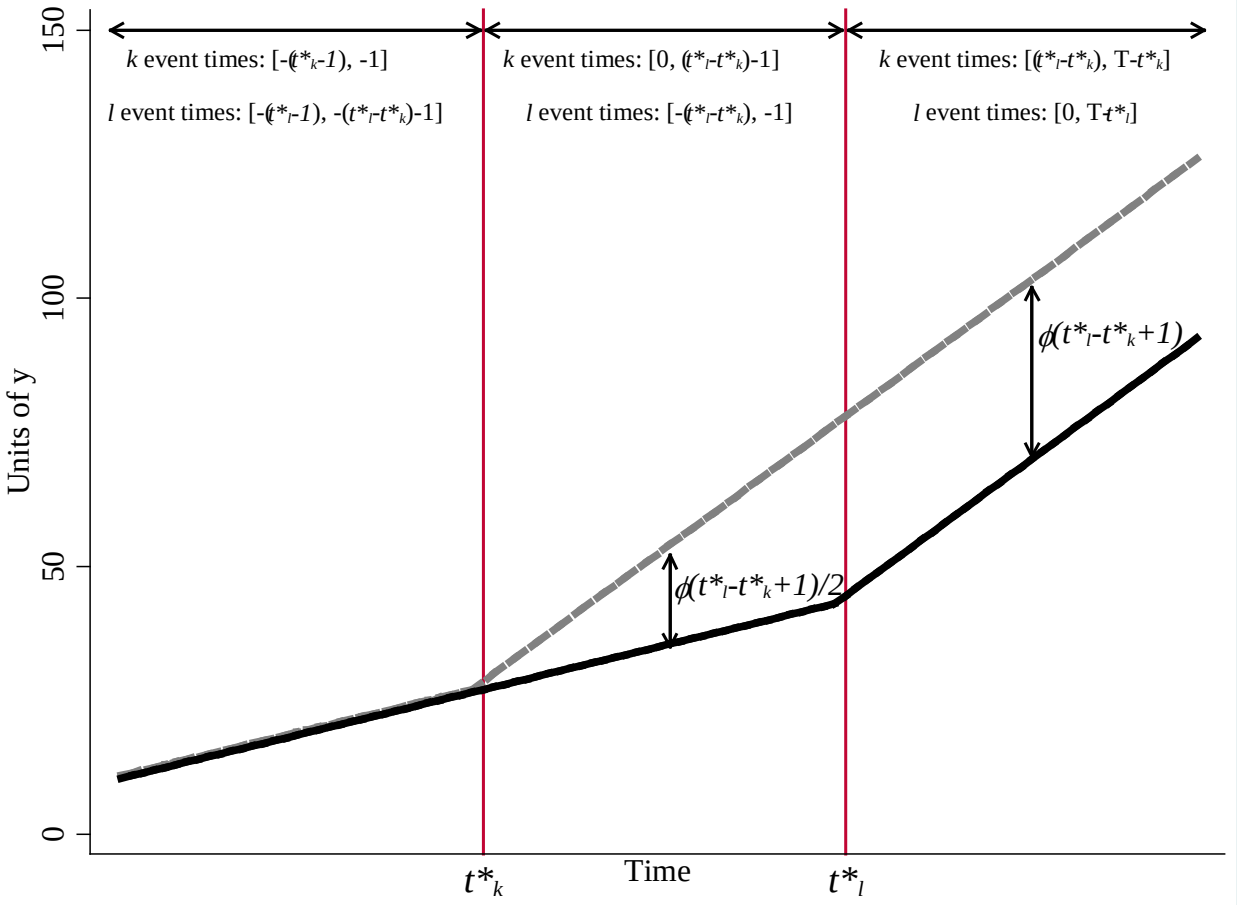
Notes: The figure plots outcomes in three groups: a control group, U , which is never treated; an early treatment group, E , which receives a binary treatment at $t_k^* = \frac{34}{100}T$; and a late treatment group, ℓ , which receives the binary treatment at $t_\ell^* = \frac{85}{100}T$. The x-axis notes the three sub-periods: the pre-period for group k , $[1, t_k^* - 1]$, denoted by $PRE(k)$; the middle period when group k is treated and group ℓ is not, $[t_k^*, t_\ell^* - 1]$, denoted by $MID(k, \ell)$; and the post-period for group ℓ , $[t_\ell^*, T]$, denoted by $POST(\ell)$. I set the treatment effect to 10 in group k and 15 in group ℓ .

Figure 2. The Four Simple (2x2) Difference-in-Differences Estimates from the Three Group Case



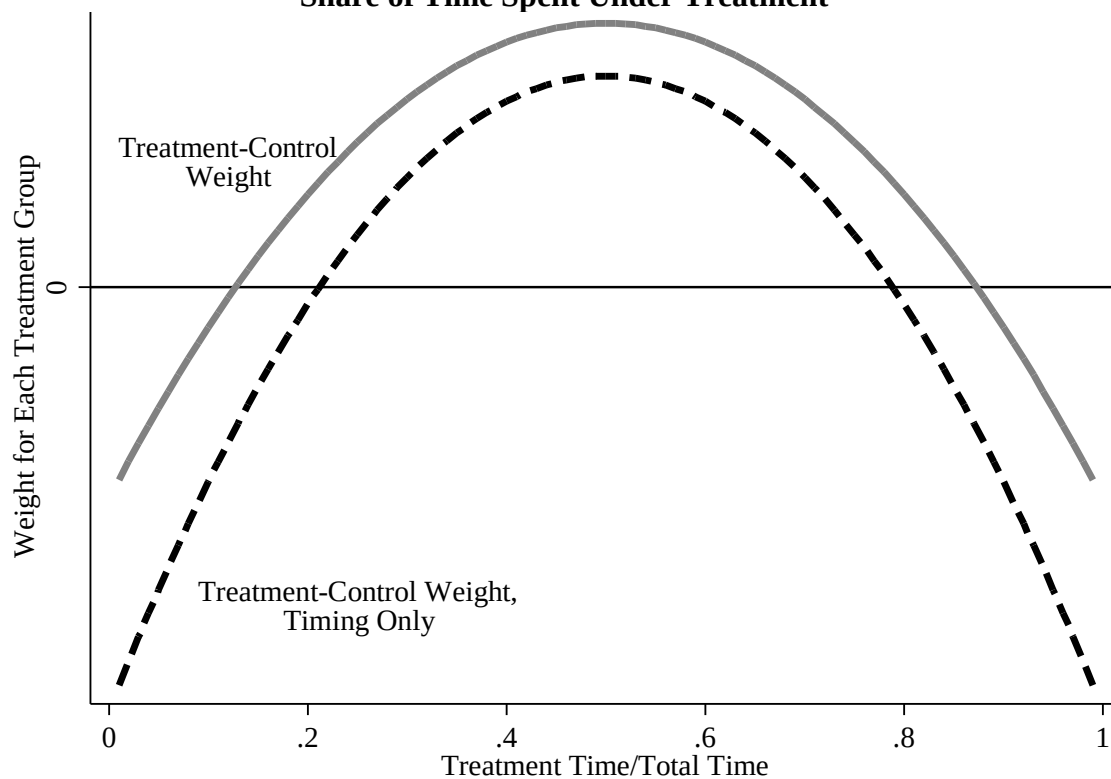
Notes: The figure plots the groups and time periods that generate the four simple 2x2 difference-in-difference estimates in the case with an early treatment group, a late treatment group, and an untreated group from Figure 1. Each panel plots the data structure for one 2x2 DD. Panel A compares early treated units to untreated units ($\hat{\beta}_{kU}^{DD}$); panel B compares late treated units to untreated units ($\hat{\beta}_{lU}^{DD}$); panel C compares early treated units to late treated units during the late group's pre-period ($\hat{\beta}_{kl}^{DD,k}$); panel D compares late treated units to early treated units during the early group's post-period ($\hat{\beta}_{kl}^{DD,l}$). The treatment times mean that $\bar{D}_k = 0.67$ and $\bar{D}_l = 0.16$, so with equal group sizes, the decomposition weights on the 2x2 estimate from each panel are 0.365 for panel A, 0.222 for panel B, 0.278 for panel C, and 0.135 for panel D.

Figure 3. Difference-in-Differences Estimates with Variation in Timing Are Biased When Treatment Effects Vary Over Time



Notes: The figure plots a stylized example of a timing-only DD set up with a treatment effect that is a trend-break rather than a level shift (see Meer and West 2013). Following section II.A.ii, the trend-break effect equals $\phi \cdot (t - t^* + 1)$. The top of the figure notes which event-times lie in the $PRE(k)$, $MID(k, \ell)$, and $POST(\ell)$ periods for each unit. The figure also notes the average difference between groups in each of these periods. In the $MID(k, \ell)$ period, outcomes differ by $\frac{\phi}{2}(t_l^* - t_k^* + 1)$ on average. In the $POST(\ell)$ period, however, outcomes had already been growing in the early group for $t_l^* - t_k^*$ periods, and so they differ by $\phi(t_l^* - t_k^* + 1)$ on average. The 2x2 DD that compares the later group to the earlier group is biased and, in the linear trend-break case, weakly negative despite a positive and growing treatment effect.

Figure 4. Weighted Common Trends: The Treatment/Control Weights as a Function of the Share of Time Spent Under Treatment



Notes: The figure plots the weights that determine each timing group's importance in the weighted common trends expression in equations (16) and (17).